DETAILED PROJECT REPORT

Predictive Forecasting of Care Load & Placement Demand

Integrated project report and detailed research paper

Interactive forecasting and decision-support dashboard
Exploratory data analysis • Time-series forecasting • Flow analysis • Capacity scenarios

PROTOTYPE STUDY — SIMULATED DATA ONLY

Numerical examples and KPI values are illustrative, not official HHS statistics, and are not suitable for operational decisions.

Report Information

Item	Description
Project domain	Care-load forecasting and placement planning
Application	Python and Streamlit interactive dashboard
Forecast scope	Daily trend analysis and short-term forecasts
Analytical methods	EDA, time-series forecasting, flow balance, baseline comparison, capacity scenarios
Report status	Integrated detailed prototype report; simulated data

Data Disclaimer

The dashboard currently generates simulated data without CSV upload. Numerical examples, forecasts and KPI values in this report are illustrative, not official HHS statistics, and must not be used for operational decisions.

Executive Summary

This project develops an interactive decision-support dashboard to examine child-care workload and placement demand. Its primary purpose is to forecast the number of children in HHS care, estimate the balance between intake and exits, and predict short-term discharge demand. The dashboard combines trend exploration, approximate forecast ranges, capacity-threshold scenarios, and comparison of simple forecasting baselines.

The application is implemented using Python, Streamlit, Pandas, NumPy, Matplotlib, and scikit-learn. Users can select the analysis period, target measure, smoothing window, forecast horizon, and a scenario threshold. The dashboard presents point forecasts with uncertainty indications and model-error measures.

The present application is a demonstrable prototype. Simulated data illustrate the workflow rather than actual demand levels or validated model performance. Operational adoption requires authorized historical data, consistent definitions, stock-flow reconciliation where applicable, stakeholder-approved KPIs and capacity thresholds, repeated out-of-sample validation, and governance review.

1. Introduction

Planning for care and placement demand requires visibility into both the number of children currently in care and the flow of children entering, transferring between, or exiting care. A stock measure such as the number of children in care represents accumulated prior activity at a point in time. Flow measures such as intake and discharge describe movement during a period. Reviewing these measures together can help identify whether workload is stable, increasing, or decreasing.

Descriptive dashboards summarize historical activity, while a planning workflow also benefits from estimates of what may happen over a short future horizon. Forecasts are uncertain and may be affected by policy changes, operational disruptions, data-quality issues, and shifts in underlying patterns. Accordingly, forecasts should be presented with validation metrics, uncertainty information, and limitations.

This project brings time-series exploration, simple forecasting, model comparison, and capacity scenarios into one interactive interface. The focus is an interpretable prototype that can later be evaluated using authorized historical data.

2. Problem Statement

The project addresses the limited advance visibility that may result when care workload is reviewed only through current counts or isolated reports. A decision-support application is needed to organize daily measures, summarize trends, estimate near-term care load and discharge demand, and flag when a forecast approaches a specified planning threshold. The application must make clear that estimates are uncertain and that capacity thresholds and KPI definitions require operational approval.

3. Research Questions

1. How can the number of children in HHS care be represented as a short-term time-series forecast?
2. How can the difference between simulated intake and exits be summarized to indicate a possible workload imbalance?
3. What short-term discharge volume is suggested by the selected forecasting method?
4. How can a forecast threshold crossing be communicated as an early-warning signal?
5. How should forecast error and uncertainty be represented so users do not over-interpret a point estimate?
6. How can statistical and machine-learning approaches be compared fairly in a later validated study?

4. Objectives and Scope

4.1 Primary Objectives

- Forecast the number of children in HHS care.
- Estimate future imbalance between intake and exits.
- Predict short-term discharge demand.

4.2 Secondary Objectives

- Provide early warnings for healthcare planners.
- Quantify forecast uncertainty.
- Compare statistical versus machine-learning forecasting approaches.

4.3 In-Scope Deliverables

- Interactive Streamlit dashboard with trend and forecast views.
- EDA-oriented summaries and charts.
- Forecast output with approximate lower and upper estimates.
- Flow-balance and capacity-threshold scenario views.
- Model comparison and KPI interpretation.
- Detailed project report / research paper for stakeholder review.

5. Data Description and Synthetic Data Design

The current demonstration produces approximately 1,095 daily observations, corresponding to roughly three years. A fixed random seed is used in the generation routine so that the demonstration is reproducible in the same software environment. The data are synthetic and intentionally contain count variation, seasonal-like components, and gradual changes to make visual analytics meaningful.

Field / measure	Analytical role	Data status
Date	Daily time index used for plotting, filtering, training and forecasting.	Generated date sequence
Children apprehended and placed in CBP custody	Daily intake-like count.	Simulated
Children in CBP custody	Stock count of children in CBP custody.	Simulated recursive stock
Children transferred out of CBP custody	Daily transfer-out count.	Simulated
Children in HHS Care	Primary care-load stock measure.	Simulated recursive stock
Children discharged from HHS Care	Daily discharge / exit count.	Simulated
Simulated HHS inflow	Flow series used in inflow-versus-discharge comparison.	Simulated separately

The simulated inflow series is generated separately and is not fully reconciled to the simulated HHS care stock. A production dataset should establish a consistent stock-flow identity where applicable, including opening stock, entries, exits, transfers, and documented adjustments. Field definitions, reporting cut-offs, and whether counts represent daily events or end-of-day stocks must be documented before modelling.

6. Exploratory Data Analysis

6.1 Data Inspection

The dashboard provides date filtering, summary statistics, and time-series plots for selected measures. The EDA workflow should verify date continuity, uniqueness of dates, missing values, non-negative counts, and consistent measurement definitions. Operational data may require explicit handling of incomplete reporting days, duplicate rows, late corrections, and structural breaks.

6.2 Trend and Smoothing

Daily plots display short-term fluctuations. A configurable rolling mean reduces noise and exposes broad direction. The smoothing window is an analytical control and should not replace the original series during validation. Analysts should compare raw and smoothed series so that spikes are not concealed.

6.3 Distribution and Variability

Descriptive summaries such as mean, median, standard deviation, minimum, maximum, and percentiles describe typical levels and variability. For event counts, distribution review can identify unusually high or low days. In real operational data, apparent outliers should be investigated for data errors or meaningful events rather than automatically removed.

6.4 Flow Balance

The flow analysis compares inflow and discharge totals over a selected period. Net flow is defined for the demonstration as total inflow minus total discharges. Positive net flow means generated inflow exceeds generated exits; negative net flow means exits exceed inflow. This interpretation is valid only when both series use compatible definitions, periods, and populations. The prototype's synthetic inflow is independent, so its net flow is illustrative.

7. Forecasting Methodology

7.1 Forecast Target and Horizon

The user selects a target measure and future horizon, with the dashboard supporting short-range forecasts. The principal care-load target is a stock count, while discharge is a daily event count. These targets have different statistical meanings and should be evaluated separately.

7.2 Demonstration Model

The forecast module fits a linear regression against the observation index and blends the extrapolated trend with the recent 14-day average. This hybridized estimate is intended to moderate a trend that may otherwise extrapolate too aggressively. Since the variables represent counts, negative predictions are clipped at zero. This transparent approach demonstrates the interface but is not established as the best model for the real process.

7.3 Baseline Model Comparison

The current model-performance view compares simple baselines on a chronological holdout: last observed value, a 7-day moving-average baseline, and linear trend. These provide reference points for evaluating more sophisticated methods. A future study should add statistical models such as exponential smoothing or ARIMA and machine-learning models such as Random Forest or Gradient Boosting, using identical data splits and consistent tuning rules.

7.4 Validation Design

A chronological train/holdout split uses earlier observations for training and later observations for evaluation. This is preferable to randomly shuffling a time series, which can allow future observations to inform a model evaluated on the past. Operational readiness calls for repeated rolling-origin or walk-forward validation, assessing performance at multiple forecast origins and horizons.

8. Metrics and Uncertainty

Metric	Definition	Interpretation
MAE	Mean of absolute prediction errors.	Average error magnitude in the target's units.
RMSE	Square root of mean squared errors.	Penalizes larger errors more strongly than MAE.
MAPE	Mean absolute percentage error, for non-zero actual values.	Relative error; unstable when actual values are zero or small.
Accuracy proxy	100% minus MAPE, bounded to 0–100% in dashboard logic.	Project-defined summary, not a universal forecast-accuracy standard.
Approximate forecast range	Point estimate ± 1.96 residual standard deviations; lower bound clipped at zero.	Illustrative residual band, not a calibrated prediction interval.
Capacity exposure proxy	Share of forecast days where upper estimate reaches a selected threshold, where available.	Scenario indicator, not a calibrated breach probability.

A residual-based range does not automatically have reliable coverage. A production implementation should calibrate intervals on out-of-sample residuals, test coverage by horizon, and communicate the assumptions used.

9. Illustrative Findings from the Demonstration

The seeded demonstration series is constructed to show a rising workload pattern. In the generated series, the HHS care stock starts at 1,850 and ends at 3,444, a difference of 1,594 or approximately 86.2% over the generated interval. The separately generated inflow totals 26,314, while simulated discharges total 24,076, resulting in an illustrative net difference of +2,238. The average daily generated inflow is approximately 24.0 and the average daily discharge count is approximately 22.0.

These figures are outputs of the synthetic data generator. They explain how the dashboard summarizes trends and flow balance; they are not observed program statistics, do not establish a real increase in care load, and must not be used in government operational decisions.

9.1 What the Findings Demonstrate

- The interface distinguishes stock measures from daily flow measures.
- A positive simulated net flow can be communicated alongside the care-load trend.
- A forecast can be shown with a range rather than as a single certain value.
- Threshold crossing can be translated into a time-to-warning output when a valid threshold is supplied.

10. Dashboard Design and User Workflow

1. Open the dashboard and review headline indicators and the care-load time series.
2. Select the analysis period and target measure.
3. Inspect daily trends and adjust the rolling-average window.
4. Choose a forecast horizon and review the forecast trajectory and approximate range.
5. Review flow comparison and enter a scenario threshold only when it is defined for demonstration or approved for operational use.
6. Compare baseline model errors and interpret MAE/RMSE in the target's units.
7. Review objective findings, KPI outputs, interpretation, and recommended next actions at the bottom of the dashboard.

10.1 Dashboard Sections

Dashboard section	Purpose
Overview	Headline values and care-load trend; selected-period totals for apprehensions, transfers and discharges.
Trends	Measure selection, rolling-average smoothing, descriptive statistics and multi-measure comparison.
Forecast	Selected measure; 7–90 day horizon; forecast chart, approximate lower/upper estimates, holdout MAE/RMSE and CSV export.
Flow & Capacity	Illustrative inflow-versus-discharge trends and user-entered capacity-threshold scenario.
Model Performance	Chronological holdout comparison of last-value, 7-day moving average and linear-trend baselines.
Final outcomes	Objective findings, KPI proxy values and recommended planning actions.

11. KPI Definitions and Interpretation

KPI	Definition and interpretation
Forecast Accuracy (%)	Prototype proxy: 100 minus holdout MAPE, bounded to 0–100%. Interpret alongside MAE/RMSE; it is not a guarantee.
Surge Lead Time	Days from forecast origin to the first forecast day reaching the selected threshold. Requires an agreed threshold.
Capacity Exposure Proxy (%)	Share of forecast days where approximate upper estimate reaches/exceeds threshold. Not a calibrated breach probability.
Forecast Stability Proxy	Project-defined indicator based on residual variation relative to average observed discharge volume; not standardized.
Model Robustness Proxy	Project-defined score based on MAE spread across three baseline models on one holdout; not a substitute for repeated backtesting.

12. Objective-wise Solutions / Expected Answers

Objective 1 — Forecast Care Load

The dashboard generates a daily forecast for the selected 7–90 day horizon and displays point estimates with approximate lower/upper ranges. The horizon average and trajectory provide the expected output. Actual conclusions require validated historical data.

Objective 2 — Estimate Intake/Exit Imbalance

The flow view compares inflow and discharge totals and computes net flow as inflow minus exits. Positive net flow indicates entries exceed exits during the selected period; negative net flow indicates exits exceed inflow. In this prototype, inflow is independently simulated.

Objective 3 — Predict Discharge Demand

The discharge forecast estimates daily discharge counts and sums them over the selected horizon. This can inform planning as one signal, subject to validation against actual discharge records.

Secondary Objective — Early Warnings

The dashboard identifies the first forecast day on which the care-load point forecast reaches the selected threshold. Upper estimates can support scenario review. The threshold must be approved and operationally defined.

Secondary Objective — Quantify Uncertainty

The forecast chart presents a residual-based approximate range to communicate uncertainty. Real deployment requires interval calibration and repeated out-of-sample evaluation.

Secondary Objective — Compare Statistical and ML Approaches

The current prototype compares simple baselines only. A complete comparison should add statistical and machine-learning candidates and evaluate them on identical chronological folds using agreed metrics.

13. Recommended Planning Actions

1. Review forecasted care load and the upper estimate against approved capacity limits.
2. Use discharge forecasts as one input to staffing and placement planning, not as a stand-alone decision.
3. Monitor sustained positive net flow because it may contribute to rising care load.
4. Investigate forecast threshold crossings and validate them with current operational information.
5. Review model errors regularly and recalibrate when patterns change.
6. Document definitions for intake, transfer, discharge, capacity, surge and forecast accuracy.

14. Discussion

The prototype demonstrates an end-to-end analytical pattern: data preparation, EDA, forecast generation, evaluation, scenario review, and communication through a dashboard. Its strength is the accessibility of the workflow and use of simple reference models. The central limitation is that synthetic data cannot establish the actual data-generating process or validate forecasting accuracy for the target environment. Performance on simulated data is not evidence that a model will perform well when policies, reporting practices, and operational constraints change.

15. Limitations and Ethical / Governance Considerations

- Data limitation: all current observations are simulated and not official statistics.
- Definition limitation: simulated flow and stock measures are not fully reconciled.
- Model limitation: the demonstration model is simple and may not capture complex seasonality, abrupt changes, or external drivers.
- Uncertainty limitation: the residual band is approximate and not calibrated.
- KPI limitation: stability, robustness, and accuracy summaries are prototype indicators and require stakeholder agreement.
- Governance: real child-related operational data may be sensitive; use access controls, data minimization, audit logging, and authorized processing.
- Decision limitation: forecasts should support planning, not replace professional judgment or established safeguards.

16. Future Work and Implementation Roadmap

1. Build a governed data-ingestion and validation pipeline using authorized historical daily records; document source, owner, extraction date and permitted use.
2. Agree on definitions for care stock, intake, transfer, discharge and reporting date; reconcile stock-flow accounting where possible and document exceptions.
3. Add diagnostics for missing dates, duplicates, invalid or negative counts, outliers, and definition changes.
4. Define capacity and surge thresholds with authorized operational owners; avoid arbitrary thresholds.
5. Run walk-forward validation across multiple forecast origins and evaluate each required horizon.

6. Compare naïve, moving-average, exponential-smoothing/ARIMA, and machine-learning methods on identical folds.
7. Report MAE and RMSE with a stakeholder-approved accuracy definition; explain MAPE limitations.
8. Calibrate forecast intervals and test empirical coverage before interpreting threshold exposure probabilistically.
9. Monitor data drift and model error after deployment, with review and rollback procedures.
10. Conduct stakeholder usability review; add dated forecast snapshots and evaluation-report exports.
11. Maintain human review and existing safeguards; do not automate individual placement decisions from forecasts.

17. Conclusion

The project provides an interactive prototype for exploring care-load trends, forecasting near-term workload and discharge demand, reviewing flow balance, and testing capacity scenarios. It converts project objectives into visible analytical outputs and a dashboard workflow. The present evidence is demonstrative because the dataset is synthetic. The next stage is a controlled validation exercise using authorized historical data, agreed definitions, robust backtesting, calibrated uncertainty, and governance review before any operational adoption.

Appendix A — Technical Stack

- Python — application and modelling logic.
- Streamlit — interactive dashboard interface and controls.
- Pandas and NumPy — data generation, manipulation and numerical calculations.
- Matplotlib — time-series and comparison charts.
- scikit-learn — linear regression and forecast error metrics.

Appendix B — Proposed Operational Data Readiness Checklist

- Document source system, data owner, extraction date, and permitted use.
- Confirm one row per date or define aggregation logic.
- Check completeness, duplicates, non-negative values, and data types.
- Document changes in reporting definitions and major policy events.
- Confirm whether counts are daily flows or point-in-time stocks.
- Approve capacity and surge thresholds.
- Define validation windows, forecast horizon, and acceptance criteria.

Appendix C — Terminology

Term	Meaning in this report
Care-load stock	Number of children in HHS care at a point in time.
Flow / event count	Number of entries, transfers, or discharges during a defined period.
Net flow	Illustrative total inflow minus total discharges for the selected period.
Forecast horizon	Number of future days for which estimates are generated.
Holdout	Later chronological observations reserved for model evaluation.
Capacity threshold	Scenario boundary used to flag potential workload pressure; requires operational approval.